UNIT 1 · TOPIC 6(A)

Harvesting Equipment — Reapers and Combines

Syllabus topic: Equipment for harvesting, threshing, mowing, chaff cutting and baling

Reapers and windrowers

- **Vertical conveyor reaper windrower (VCR):** a cutter bar + star wheels + vertical conveyor belts lay the cut crop in a windrow to one side. **3–5 hp** walk-behind (paddy/wheat); **0.15–0.2 ha/h**.
- **Reaper-binder:** reaps and ties bundles with twine.
- **Swather/windrower:** large-area cereal and forage harvest.

Combine harvester (Kepner ch. 17)

A combine does every step in one machine:

1. **Cut and gather** — header, reel, cutter bar, auger.
2. **Feed** — feeder conveyor.
3. **Thresh** — cylinder–concave.
4. **Separate** — straw walkers or rotors.
5. **Clean** — chaffer, sieve, cleaning fan.
6. **Handle grain** — elevators, grain tank.

Settings

- **Reel index** (reel peripheral speed ÷ forward speed) = **1.25–1.5**. Place the reel axis ahead of the cutter bar, at about the height of the crop's centre of gravity.
- **Cylinder–concave clearance:** more at entry, less at exit — wheat **~10–20 mm** front, **3–10 mm** rear (adjust per crop).

Types

- Tractor-drawn (PTO).
- Self-propelled wheat/paddy combines — most common in India, **3.6–4.3 m** cutter bar.
- Track combines for wet paddy fields.

Losses and residue

- **Header loss** (shatter + cutter bar + lodged) is usually the largest loss; measure it with the **frame-count method**.
- **Super SMS** (straw management system) chops and spreads straw behind the combine — a mandatory fitment on combines in Punjab/Haryana for residue management.

Practice MCQs

Q56. Reel index (reel speed ÷ forward speed) of a combine is normally:

(a) 0.8–1.0
(b) 1.25–1.5
(c) 2.5–3.0
(d) 4.0–5.0

Answers

Q56 — (b) 1.25–1.5